

What Is Output?

- **Output** is data that has been processed into a useful form



What Is Output?

- An **output device** is any type of hardware component that conveys information to one or more people

Display devices

Printers

Speakers,
headphones,
and earbuds

Data projectors

Interactive
whiteboards

Force-feedback
game
controllers

Tactile output

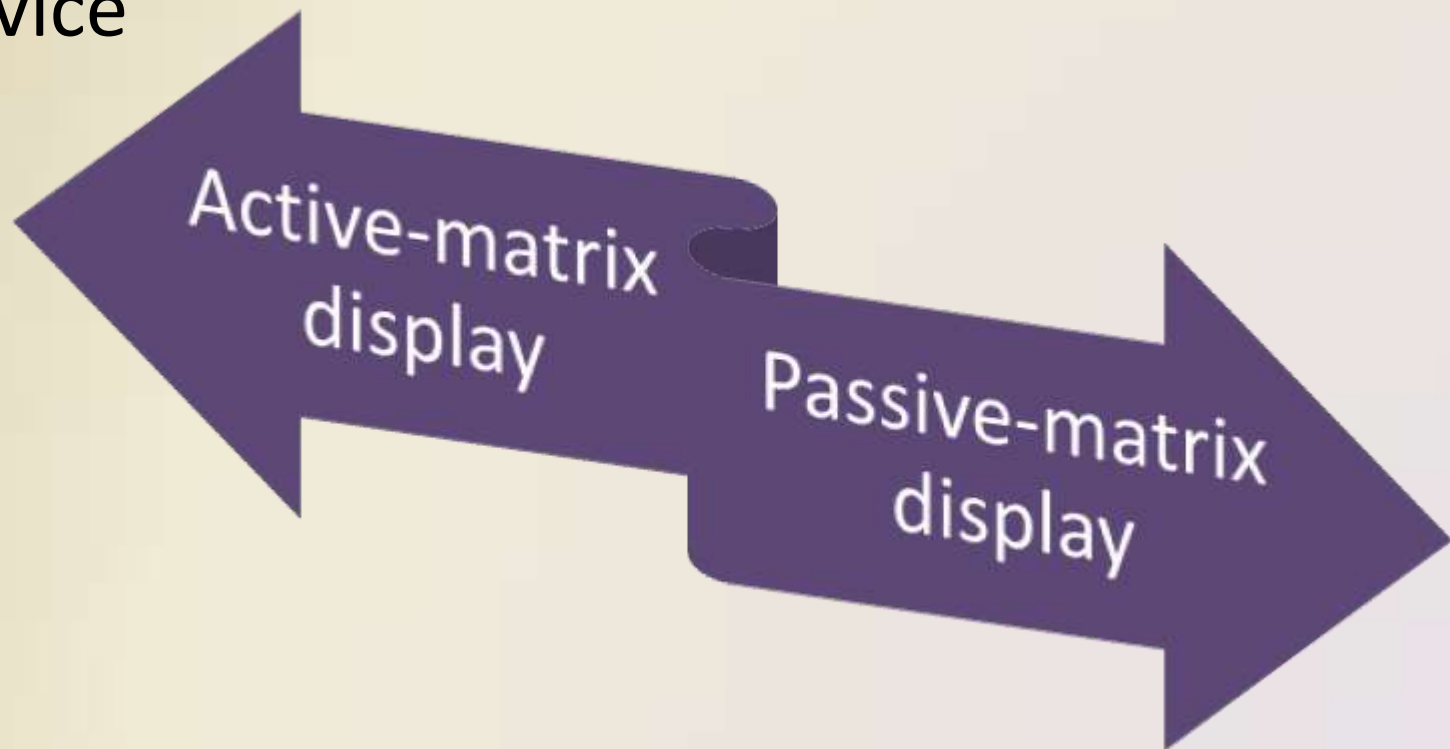
Display Devices

- A **display device** visually conveys text, graphics, and video information
- A **monitor** is packaged as a separate peripheral
 - **LCD monitor**
 - Widescreen



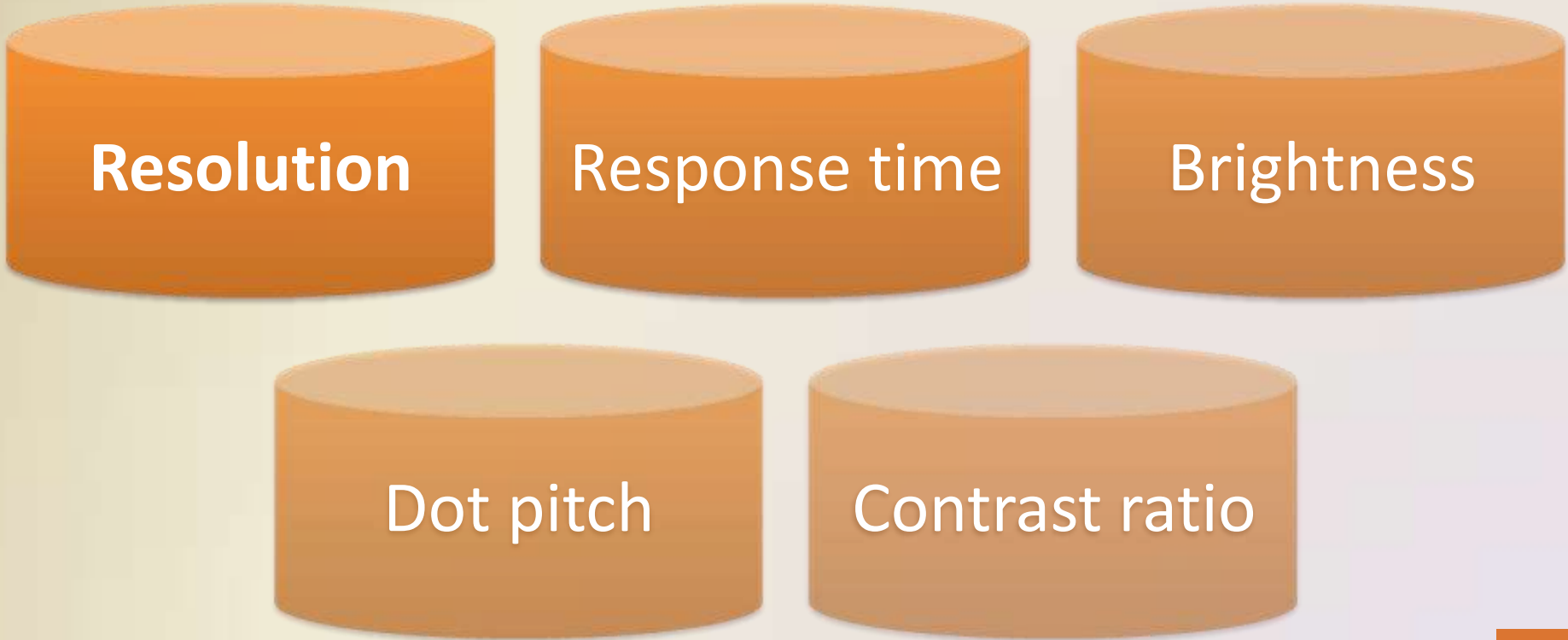
Display Devices

- **Liquid crystal display (LCD)** uses a liquid compound to present information on a display device



Display Devices

- The quality of an LCD monitor or LCD screen depends primarily on:



Resolution

Response time

Brightness

Dot pitch

Contrast ratio

Display Devices

- **Resolution** is the number of horizontal and vertical pixels in a display device
 - A higher resolution uses a greater number of pixels

(screen resolution at 1024 × 768)

fewer Ribbon commands are displayed at a lower resolution

fewer spreadsheet cells are displayed at a lower resolution (only columns A through I are displayed)

(screen resolution at 1366 × 768)

more Ribbon commands are displayed at higher resolution

more spreadsheet cells are displayed at a higher resolution (columns A through L are displayed)

(screen resolution settings)

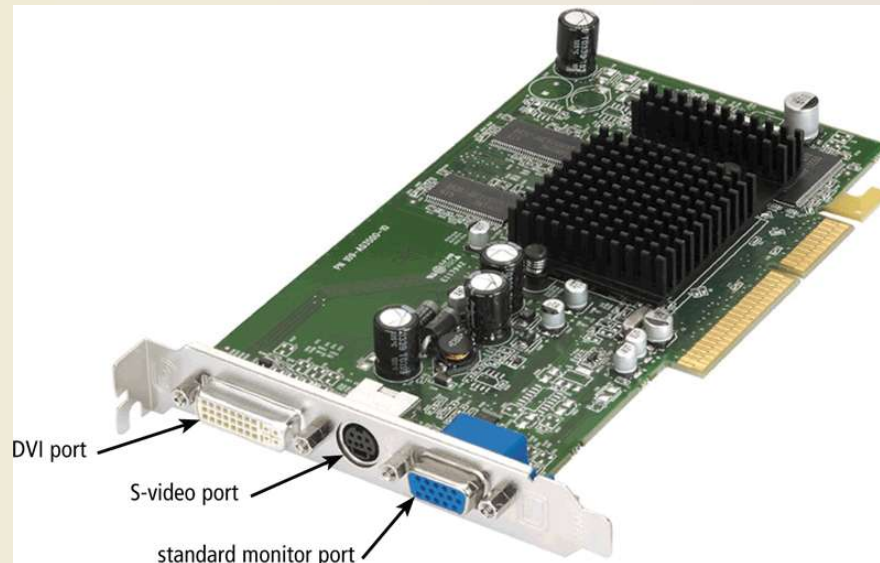
current resolution

each monitor connected to a system unit can display a different resolution

The collage consists of three main parts. The top part shows a screenshot of a 'Loan Payment Calculator' spreadsheet at 1024x768 resolution. A text box points to the Ribbon, stating 'fewer Ribbon commands are displayed at a lower resolution'. Another text box points to the spreadsheet, stating 'fewer spreadsheet cells are displayed at a lower resolution (only columns A through I are displayed)'. The middle part shows the same spreadsheet at 1366x768 resolution. A text box points to the Ribbon, stating 'more Ribbon commands are displayed at higher resolution'. Another text box points to the spreadsheet, stating 'more spreadsheet cells are displayed at a higher resolution (columns A through L are displayed)'. The bottom part shows the 'Change the appearance of your displays' control panel window. It shows two displays, with the first display set to 1280x768 resolution. A text box points to the resolution dropdown, stating 'current resolution'. Another text box points to the display icons, stating 'each monitor connected to a system unit can display a different resolution'.

Display Devices

- The graphics processing unit (GPU) controls the manipulation and display of graphics on a display device
- LCD monitors use a digital signal and should plug into a DVI port, an HDMI port, or a DisplayPort



Display Devices

- **Plasma monitors** are display devices that use gas plasma technology and offer screen sizes up to 150 inches



Display Devices

- Televisions also are a good output device
 - Require a converter if you are connecting your computer to an analog television
- Digital television (DTV) offers a crisper, higher-quality output
- **HDTV** is the most advanced form of digital television



Display Devices

- A **CRT monitor** is a desktop monitor that contains a cathode-ray tube
 - Have a much larger footprint than do LCD monitors



Printers

- A **printer** produces text and graphics on a physical medium
 - Printed information is called a hard copy, or printout
 - Landscape or portrait orientation

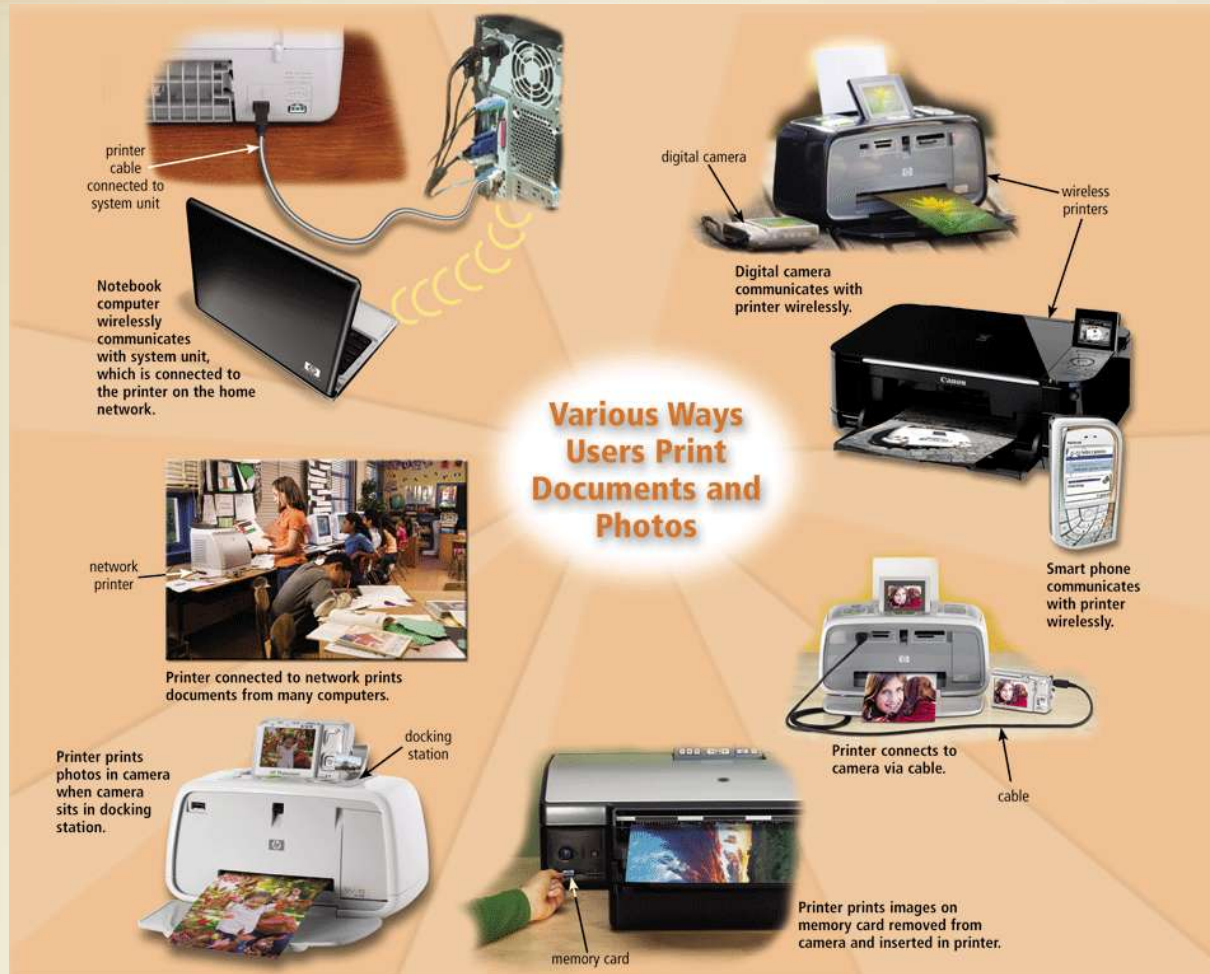


Printers

Before purchasing a printer, ask yourself a series of questions:

1. What is my budget?
2. How fast must my printer print?
3. Do I need a color printer?
4. What is the cost per page for printing?
5. Do I need multiple copies of documents?
6. Will I print graphics?
7. Do I want to print photos?
8. Do I want to print directly from a memory card?
9. What types of paper does the printer use?
10. What sizes of paper does the printer accept?
11. Do I want to print on both sides of the paper?
12. How much paper can the printer tray hold?
13. Will the printer work with my computer and software?
14. How much do supplies such as ink, toner, and paper cost?
15. Can the printer print on envelopes?
16. How many envelopes can the printer print at a time?
17. How much do I print now, and how much will I be printing in a year or two?
18. Will the printer be connected to a network?
19. Do I want wireless printing capability?

Printers



Printers

- A **nonimpact printer** forms characters and graphics on a piece of paper without actually striking the paper

Ink-jet
printers

Photo
printers

Laser
printers

Thermal
printers

Mobile
printers

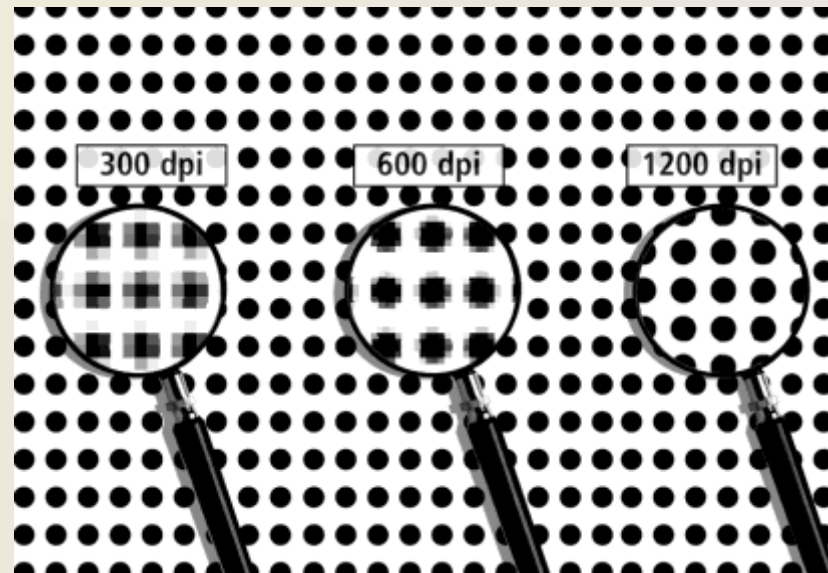
Label and
postage
printers

Plotters

Large-
format
printers

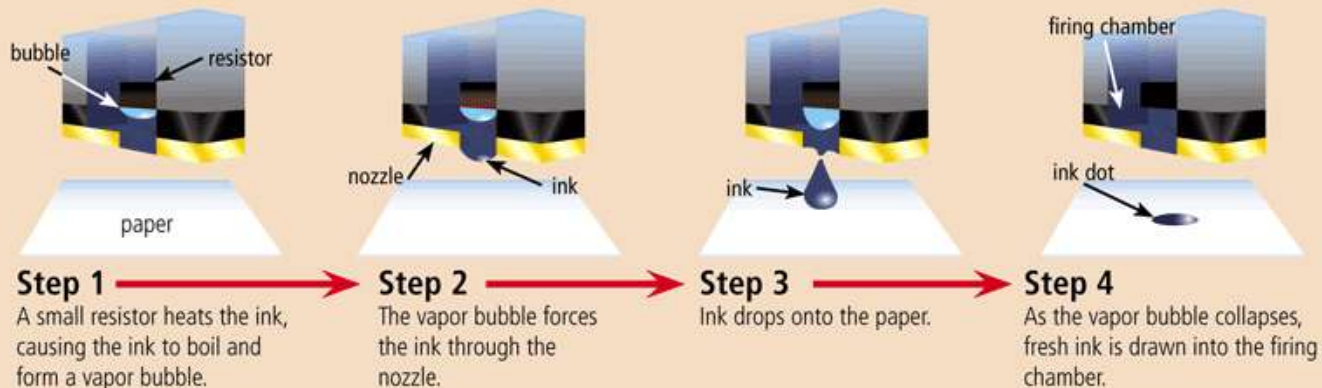
Printers

- An **ink-jet printer** forms characters and graphics by spraying tiny drops of liquid ink onto a piece of paper
 - Color or black-and-white
 - Printers with a higher dpi (dots per inch) produce a higher quality output



Printers

How an Ink-Jet Printer Works



Printers

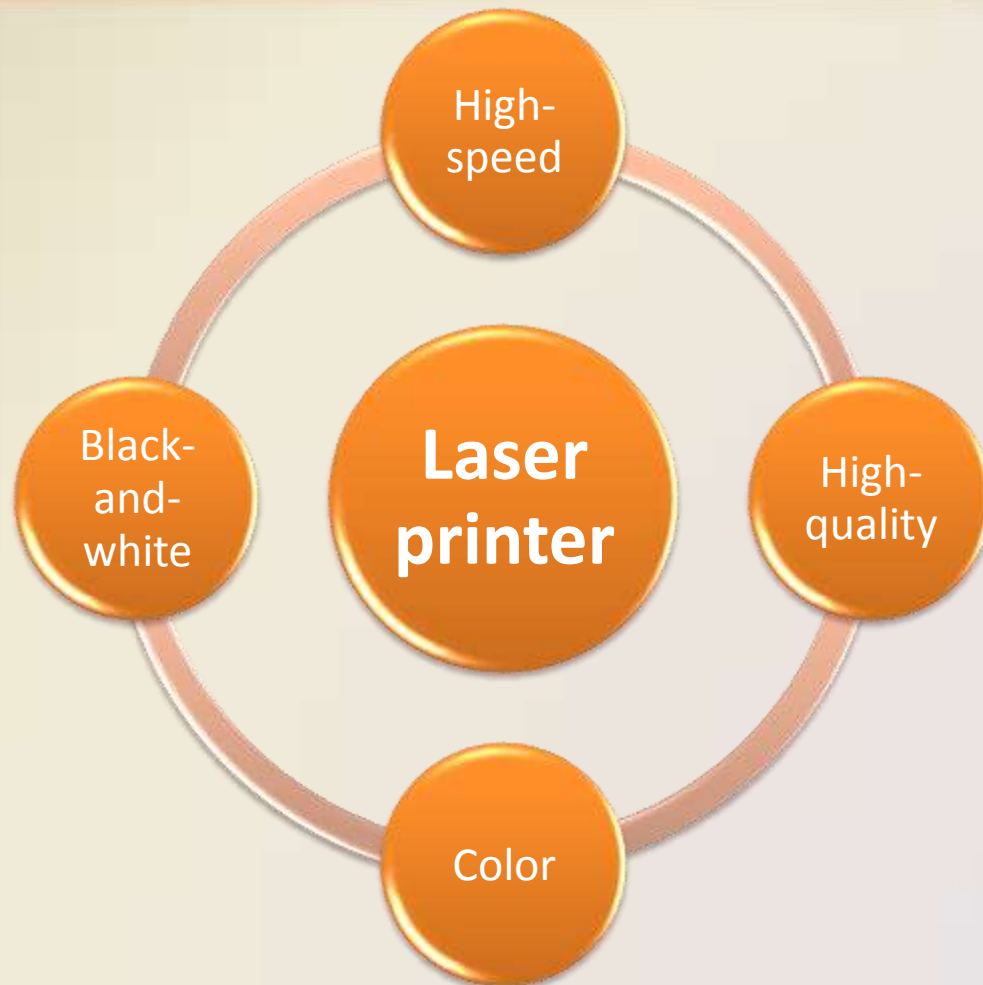
A photo printer produces color photo-lab-quality pictures

- Most use ink-jet technology
- PictBridge allows you to print photos directly from a digital camera
- Print from a memory card and preview photos on a built-in LCD screen

Printers

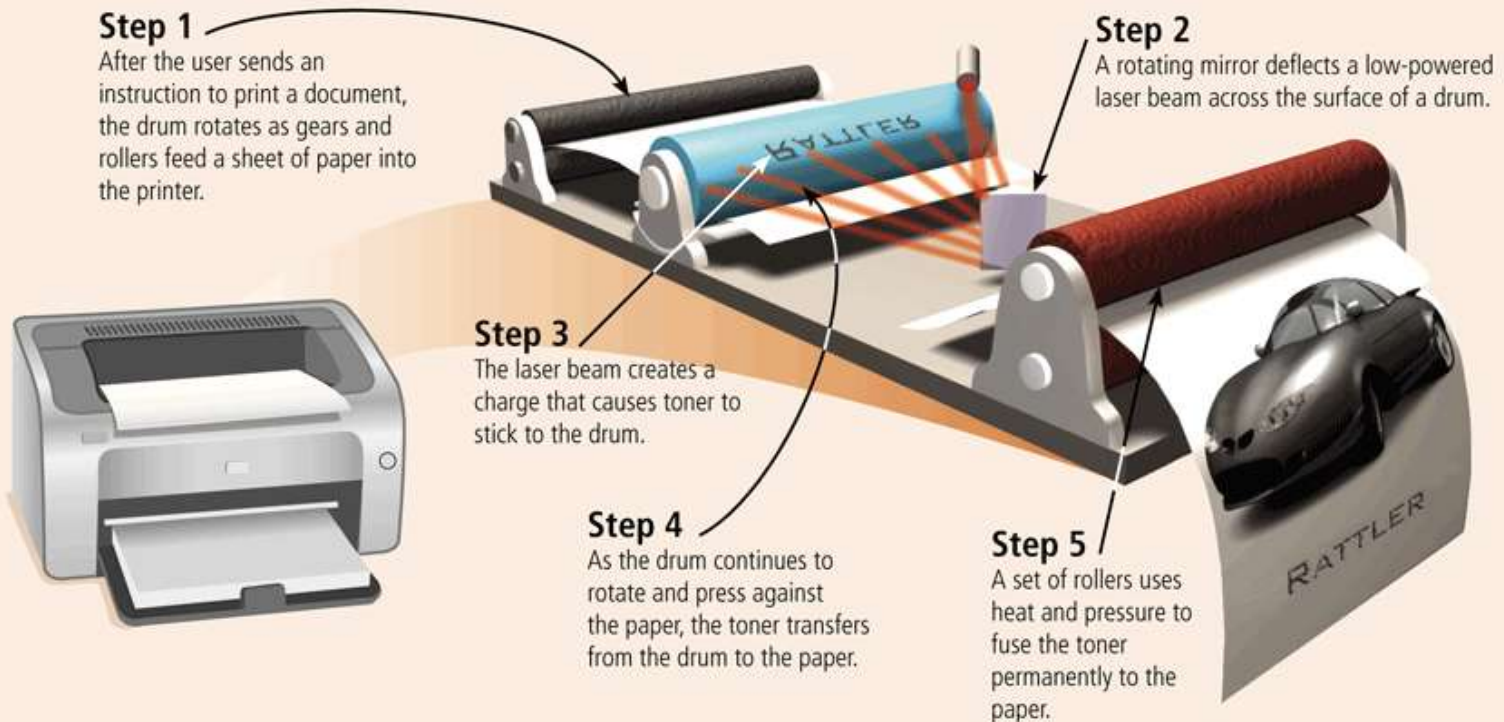


Printers



Printers

How a Black-and-White Laser Printer Works



Printers

- A **multifunction peripheral** (MFP) is a single device that prints, scans, copies, and in some cases, faxes
 - Sometimes called an all-in-one device



Printers

- A **thermal printer** generates images by pushing electrically heated pins against the heat-sensitive paper

Thermal wax-transfer
printer

Dye-sublimation
printer



Printers

- A **mobile printer** is a small, lightweight, battery-powered printer that allows a mobile user to print from a notebook computer, smart phone, or other mobile device



Printers

- A **label printer** is a small printer that prints on adhesive-type material
- A postage printer prints postage stamps
 - Postage also can be printed on other types of printers



Printers

- **Plotters** are used to produce high-quality drawings
- **Large-format printers** create photo-realistic quality color prints on a larger scale



Printers

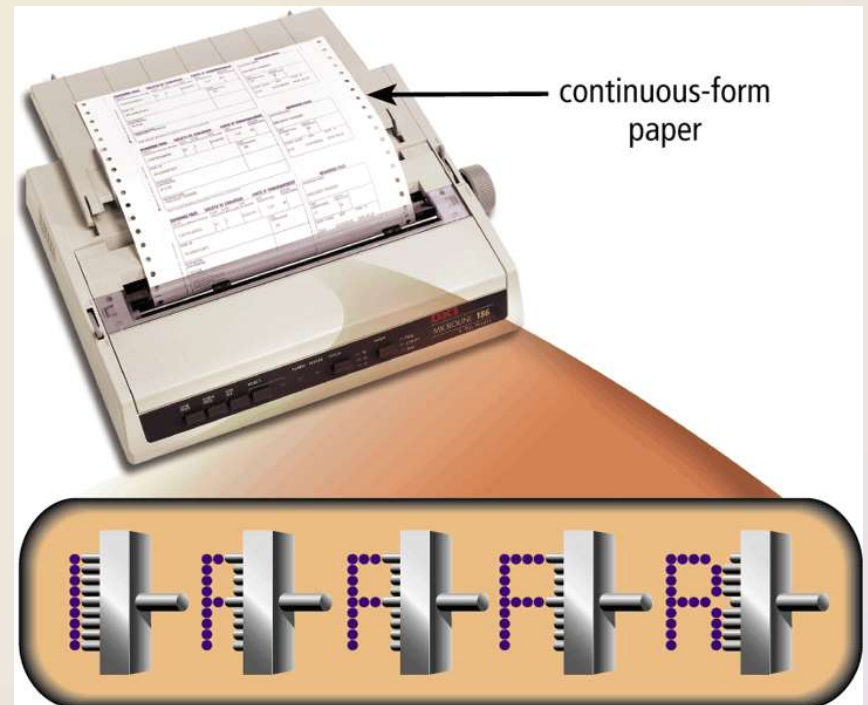
- **Impact printers** form characters and graphics on a piece of paper by striking a mechanism against an inked ribbon that physically contacts the paper

**Dot-matrix
printer**

**Line
printer**

Printers

- A **dot-matrix printer** produces printed images when tiny wire pins on a print head mechanism strike an inked ribbon
- A **line printer** prints an entire line at a time



Speakers, Headphones, and Earbuds

- An **audio output device** produces music, speech, or other sounds



Most computer users attach **speakers** to their computers to:

- Generate higher-quality sounds for playing games
- Interact with multimedia presentations
- Listen to music
- View movies

Speakers, Headphones, and Earbuds

- **Headphones** are speakers that cover or are placed outside of the ear
- **Earbuds** (also called earphones) rest inside the ear canal



Speakers, Headphones, and Earbuds

- Some speakers are specifically designed to play audio from a portable media player
 - Wireless speakers



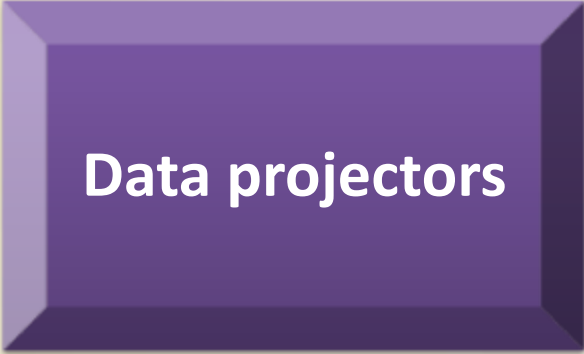
Speakers, Headphones, and Earbuds

Voice output occurs when you hear a person's voice or when the computer talks to you through the speakers

- Some Web sites dedicate themselves to providing voice output
- Often works with voice input
- VoIP uses voice output and voice input

Other Output Devices

- Other output devices are available for specific uses and applications



Data projectors



**Interactive
whiteboards**



**Force-feedback
game
controllers**



Tactile output

Other Output Devices

- A **data projector** is a device that takes the text and images displaying on a computer screen and projects them on a larger screen
 - LCD projector
 - Digital light processing (DLP) projector



Other Output Devices

- An **interactive whiteboard** is a touch-sensitive device, resembling a dry-erase board, that displays the image on a connected computer screen



Other Output Devices

- Force-feedback sends resistance to the device in response to actions of the user
- Tactile output provides the user with a physical response from the device



Putting It All Together

Suggested Output Devices by User

User

Monitor

Printer

Other

Home



- 19- or 20-inch LCD monitor, or 17-inch LCD screen on notebook computer

- Ink-jet color printer; or
- Photo printer

- Speakers
- Headphones or earbuds or headset
- Force-feedback game controller or tactile output

Small Office/Home Office



- 20- or 22-inch LCD monitor
- LCD screen smart phone or other mobile device

- Multifunction peripheral; or
- Ink-jet color printer; or
- Laser printer (black-and-white or color)
- Label printer
- Postage printer

- Speakers

Putting It All Together

Suggested Output Devices by User

User

Monitor

Printer

Other

Mobile



- 17-inch LCD screen on notebook computer
- 8.9-inch screen on a netbook
- LCD screen on smart phone or other mobile device

- Mobile color printer
- Ink-jet color printer; or
- Laser printer for in-office use (black-and-white or color)
- Photo printer

- Headphones or earbuds or headset
- DLP data projector

Power



- 30-inch LCD monitor

- Laser printer (black-and-white or color)
- Plotter or large-format printer; or
- Photo printer; or
- Dye-sublimation printer

- Speakers
- Headphones or earbuds or headset

Putting It All Together

Suggested Output Devices by User

User

Monitor

Printer

Other

Enterprise

- 20- or 22-inch LCD monitor
- LCD screen on smart phone or other mobile device

- High-speed laser printer
- Laser printer, color
- Line printer (for large reports from a mainframe)
- Label printer

- Speakers
- Headphones or earbuds or headset
- Networked DLP data projector
- Interactive whiteboard



Output Devices for Physically Challenged Users

- Many accessibility options exist for users with various disabilities

